

AI for Interpretable Heart Disease Prediction

Clinical Prediction Report

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Prediction Summary

■ Heart Disease Detected — High Risk

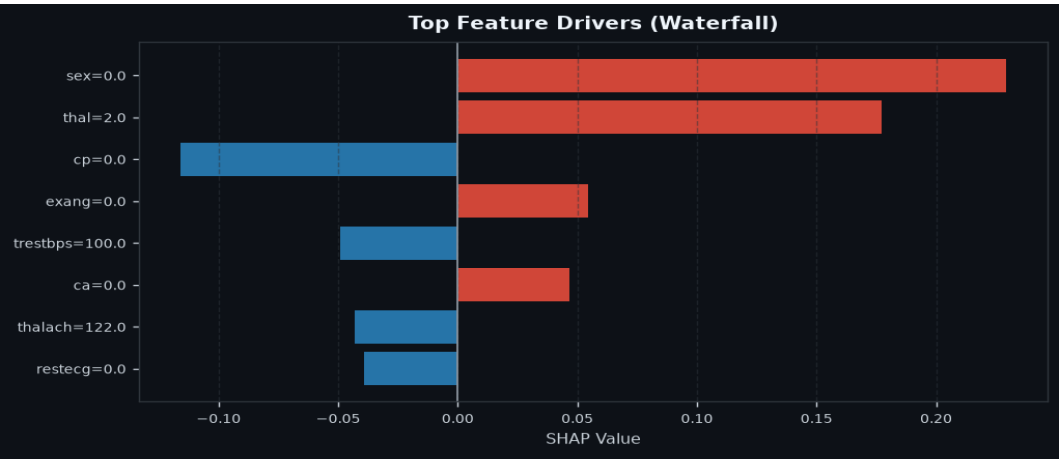
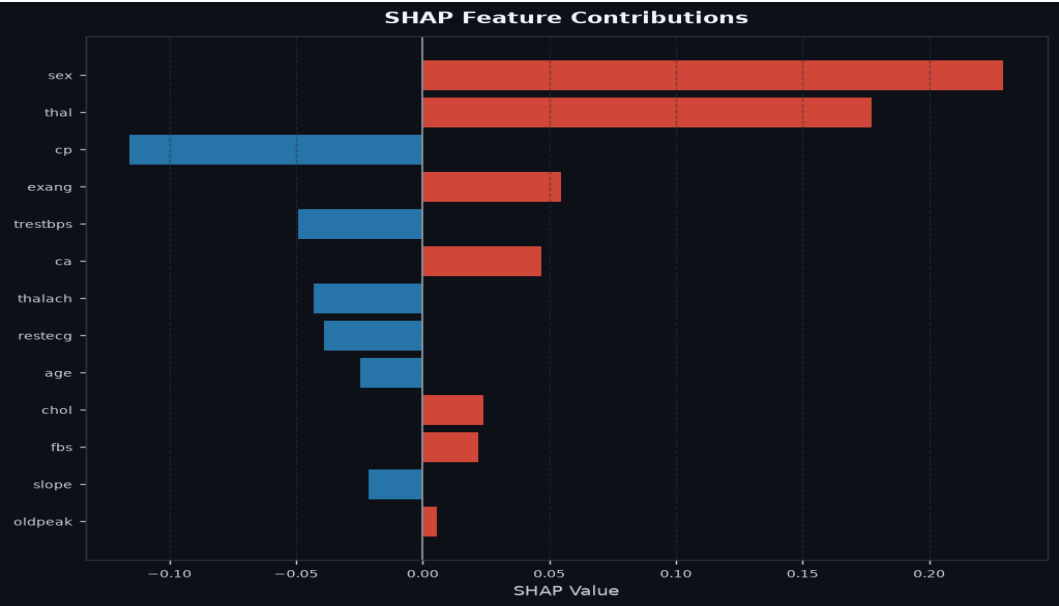
Model	P(Disease)	Weight
Random Forest	74.0%	33%
SVM (RBF)	79.1%	33%
ANN (MLP)	73.4%	33%
Ensemble (Avg)	75.5%	Final

SHAP Clinical Feature Analysis

Feature	Value	SHAP	Direction
Sex (1=Male, 0=Female)	0.0	+0.2293	↑ Risk
Thalassemia (0–3)	2.0	+0.1773	↑ Risk
Chest Pain Type (0–3)	0.0	-0.1159	↓ Risk
Exercise Induced Angina (1=Yes)	0.0	+0.0546	↑ Risk
Resting Blood Pressure (mmHg)	100.0	-0.0491	↓ Risk
Coronary Arteries Coloured (0–4)	0.0	+0.0469	↑ Risk
Max Heart Rate Achieved	122.0	-0.0431	↓ Risk
Resting ECG (0–2)	0.0	-0.0392	↓ Risk
Age (years)	58.0	-0.0247	↓ Risk
Cholesterol (mg/dl)	248.0	+0.0241	↑ Risk
Fasting Blood Sugar > 120 (1=Yes)	0.0	+0.0219	↑ Risk
Slope of ST Segment (0–2)	1.0	-0.0215	↓ Risk
ST Depression (oldpeak)	1.0	+0.0055	↑ Risk

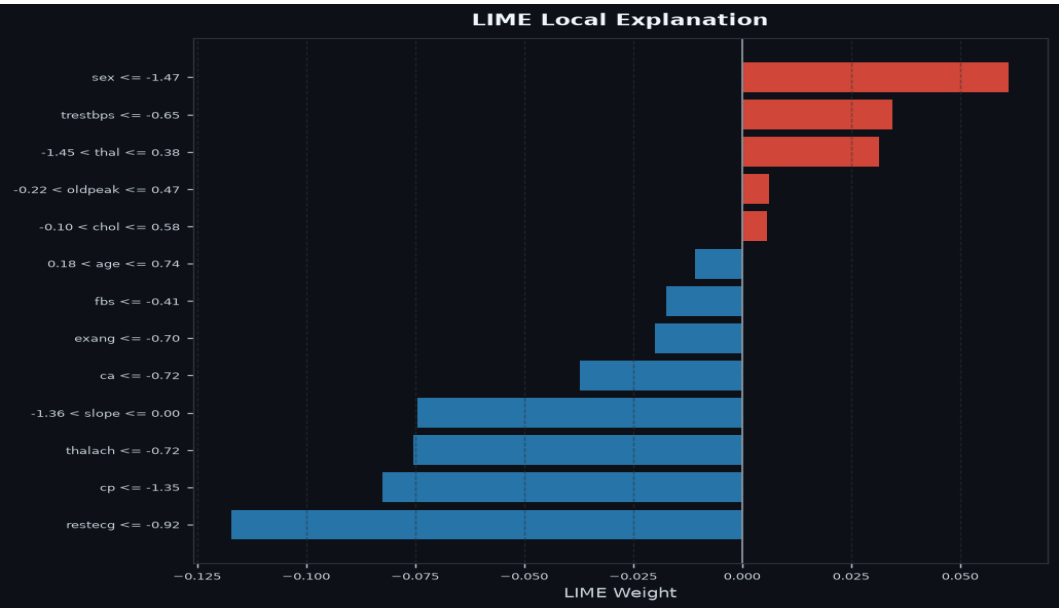
SHAP Explainability

Red bars push prediction toward disease. Blue bars push away from disease.



LIME Local Explanation

Locally fitted linear approximation around this patient's neighbourhood. Positive weights push toward disease.



■ **DISCLAIMER:** This report is generated by an AI research system for educational and research purposes only. It is NOT a substitute for professional medical advice, diagnosis, or treatment. Always consult a qualified physician.